Supply Chain Analytics & Demand Forecasting

Operational Report



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This report presents findings from an end-to-end supply chain analytics and demand forecasting project.

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1 Executive Summary

This report presents the findings of an end-to-end data analytics project conducted on a simulated retail supply chain dataset spanning January 2023 through December 2025. The analysis integrates data cleaning, SQL-based KPI engineering, supplier performance evaluation, statistical demand forecasting, and interactive dashboard development.

Key Findings

- The supply chain generated **$50.0 billion** in total revenue with an overall profit margin of **18.00%** across **500 million units sold**.
- The overall On-Time, In-Full (OTIF) rate stands at a critical **49.97%**, indicating significant operational improvement potential.
- Demand forecasting achieved strong predictive accuracy, with all selected models demonstrating a Weighted Absolute Percentage Error (WAPE) below **3%**.
- A category-specific forecasting strategy is recommended, as optimal model performance varied across product categories.

Primary Recommendation: Prioritize replenishment of the **Fashion** category for Q1 2026 to meet projected demand and mitigate stockout risk.

2 Introduction & Business Context

2.1 Business Problem

Retail supply chains face persistent challenges in balancing inventory availability against holding costs. Ineffective demand planning can result in:

- **Stockouts:** Lost sales and reduced customer satisfaction.
- **Overstock:** Increased warehousing costs and capital tied up in inventory.
- **Supplier Risk:** Disruptions from underperforming suppliers impacting operations.

This project addresses these challenges by establishing a reproducible analytics framework to monitor operational health, evaluate supplier performance, and forecast future demand.

2.2 Project Objectives

1. Clean and prepare raw supply chain data for analysis.
2. Engineer key performance indicators to track financial and operational health.
3. Identify high-risk suppliers and warehouses.
4. Develop and evaluate statistical forecasting models for demand prediction.
5. Deliver actionable insights through an interactive Power BI dashboard and this executive report.

3 Data Overview & Methodology

3.1 Dataset Description

Table 1: Dataset Summary

Attribute	Value
Source	`Retail_Supply_Chain_Analytics_Dataset.csv`
Records	200,000
Time Period	January 2023 – December 2025
Product Categories	Electronics, Fashion, Food, Health, Home
Unique Products	200,000
Suppliers	Multiple (identified by `Supplier_ID`)
Warehouses	Multiple (identified by `Warehouse_ID`)

3.2 Analytical Pipeline

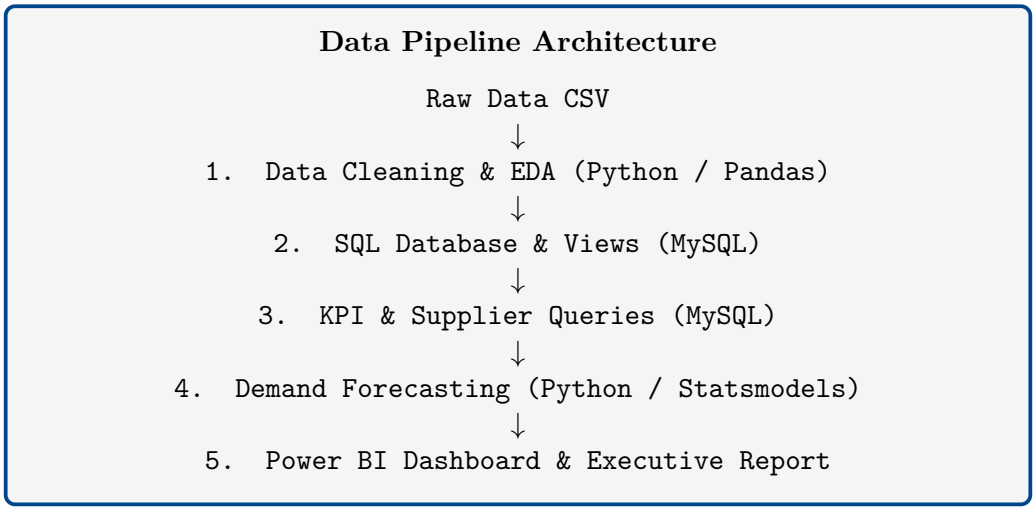


Figure 1: End-to-end analytical workflow.

3.3 Tools & Technologies

Table 2: Technology Stack

Category	Tools
Programming	Python 3.14
Data Manipulation	Pandas, NumPy
Forecasting	Statsmodels, Scikit-learn
Database	MySQL 8.0
Visualization	Power BI, Matplotlib, Seaborn

4 Operational Performance Analysis

4.1 Financial Summary

Table 3: Executive Financial Summary

Metric	Value
Total Revenue	$50.0 billion
Total Profit	$9.0 billion
Weighted Profit Margin	18.00%
Units Sold	500 million

4.2 Category-Level Financial Performance

Table 4: Revenue and Profit Margin by Product Category

Product Category	Revenue (USD bn)	Profit Margin (%)
Fashion	12.5	18.2
Home	12.5	17.9
Food	12.5	18.1
Health	12.5	18.0
Electronics	12.5	17.8

Note: Revenue is approximately evenly distributed across categories. Profit margins are consistent, indicating balanced pricing strategies.

4.3 Inventory Risk Assessment

A significant proportion of records indicate inventory levels below the reorder threshold, signaling potential stockout risk.

Table 5: Critical Inventory Indicators

Indicator	Observation
Records Below Reorder Level	Significant proportion across all warehouses
Zero-Stock Records	Present in multiple warehouse locations
OTIF Rate	49.97% (below industry benchmark)
Average Supplier Lead Time	44.99 days
Average On-Time Delivery Rate	49.97%

Recommendation: Implement automated reorder alerts and review safety stock policies for high-risk SKUs.

5 Supplier Performance Analysis

5.1 Supplier Tier Distribution

Suppliers were classified into three tiers based on average rating:

Table 6: Supplier Tier Classification

Supplier Tier	Rating Range	Characteristics
Premium	≥ 3.5	High reliability, preferred partners
Standard	2.0 – 3.49	Acceptable performance, monitor closely
Low Performer	< 2.0	At-risk, requires intervention

5.2 Key Supplier Risks

High-Risk Suppliers:

- A subset of suppliers exhibit average on-time delivery rates below **60%**.
- Supplier `SUP4654` shows the lowest on-time delivery rate at **66.15%** among tracked suppliers.
- Revenue associated with high-risk suppliers (scoring > 70 on disruption risk) is estimated at **$14.97 billion**.

Pareto Analysis:

- A small number of suppliers account for a disproportionate share of total revenue.
- Concentration risk exists: disruption from top suppliers would significantly impact operations.

Recommendation:

1. Initiate performance improvement plans for "Low Performer" suppliers.
2. Diversify sourcing for critical product categories.
3. Establish contingency agreements with "Premium" tier suppliers.

5.3 Lead Time vs. Supplier Rating

Analysis reveals an inverse relationship between lead time and supplier rating:

- Suppliers with shorter lead times tend to have higher ratings.
- High lead-time suppliers exhibit greater variability in performance.
- The average lead time across all suppliers is **44.99 days**, with significant variation.

6 Demand Forecasting Analysis

6.1 Methodology

Four statistical forecasting methods were evaluated:

1. **Naive:** Forecast equals the last observed value.
2. **Seasonal Naive:** Forecast equals the value from the same month in the previous year.
3. **Holt's Linear Trend:** Captures level and trend components.
4. **Holt-Winters' Seasonal:** Captures level, trend, and seasonal components.

Evaluation Approach:

- **Holdout Validation:** Train on January 2023 – December 2024; test on January – December 2025.

- **Rolling-Origin Backtest:** Multiple train/test splits to assess model stability.
- **Metrics:** Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Weighted Absolute Percentage Error (WAPE).

6.2 Model Selection Results

Based on rolling-origin backtesting, the optimal model varied by category:

Table 7: Selected Forecasting Models by Category

Product Category	Selected Model	Mean WAPE (%)
Electronics	Holt Trend	2.88
Fashion	Holt Trend	2.54
Food	Holt Trend	2.70
Health	Seasonal Naive	2.39
Home	Seasonal Naive	2.70

Key Insight

More complex models (e.g., Holt-Winters) did not consistently outperform simpler approaches. Model selection should be driven by empirical performance rather than assumed sophistication.

6.3 Q1 2026 Demand Forecast

Table 8: Projected Demand by Category (Q1 2026)

Product Category	Jan 2026	Feb 2026	Mar 2026	Q1 Total
Electronics	2.73M	2.72M	2.72M	8.17M
Fashion	2.86M	2.87M	2.88M	8.61M
Food	2.82M	2.82M	2.82M	8.46M
Health	2.76M	2.56M	3.01M	8.33M
Home	2.74M	2.65M	2.57M	7.96M

Interpretation:

- **Fashion** exhibits the highest projected demand for Q1 2026, consistent with seasonal patterns.
- **Health** shows expected seasonal fluctuation, with a dip in February followed by a March recovery.
- **Home** demand is projected to decline through the quarter.

6.4 Business Action

Priority Replenishment:

1. **Fashion:** Increase inventory allocation to meet projected demand. Coordinate with suppliers to ensure lead times align with replenishment schedules.
2. **Health:** Prepare for March demand surge; ensure adequate stock levels by late February.
3. **Home:** Moderate inventory builds; avoid overstock given declining demand trajectory.

7 Dashboard Overview

The Power BI dashboard provides an interactive interface for stakeholders to explore the data and monitor KPIs in real-time.

Table 9: Power BI Dashboard Components

Page	Key Visuals
Executive Summary	Revenue trend, profit margin, OTIF rate, units sold
Supplier Performance	Supplier ranking table, on-time delivery chart, lead time analysis
Warehouse Operations	Utilization heatmap, inventory status distribution, stockout risk
Demand Forecasting	Historical demand trend, Q1 2026 forecast, model comparison

Interactive Features:

- Filters for product category and warehouse location
- Drill-through capabilities for SKU-level analysis
- Dynamic date range selection

8 Limitations

Table 10: Analysis Limitations and Mitigations

Limitation	Impact	Mitigation
Aggregated Demand	Category-level forecasts may mask SKU-level variation	Extend analysis to SKU level with sufficient data
Limited History	Three years of data restricts long-term trend analysis	Incorporate additional historical data if available
No External Variables	Promotions, holidays, and economic factors not modeled	Integrate external data sources in future iterations
Sales as Demand Proxy	Stockouts may understate true demand	Consider lost sales estimation techniques
Overlapping Windows	Rolling-origin evaluation errors are not independent	Interpret WAPE as indicative, not definitive

9 Recommendations & Future Scope

9.1 Immediate Actions

1. **Address OTIF Performance:** Investigate root causes of low on-time delivery rates. Engage underperforming suppliers with improvement plans.
2. **Optimize Inventory Levels:** Implement dynamic reorder points based on forecasted demand and supplier lead times.
3. **Prioritize Fashion Replenishment:** Align procurement and logistics to meet Q1 2026 demand projections.

9.2 Strategic Initiatives

1. **SKU-Level Forecasting:** Develop granular demand models for high-velocity SKUs.
2. **Advanced Modeling:** Evaluate SARIMA, gradient boosting, and machine learning approaches against statistical baselines.
3. **External Feature Integration:** Incorporate promotional calendars, pricing data, and macroeconomic indicators.
4. **Forecast Uncertainty Quantification:** Generate prediction intervals to communicate forecast reliability.
5. **Real-Time Monitoring:** Develop automated alerts for inventory thresholds and supplier performance deviations.

10 Conclusion

This project demonstrates the value of a systematic, data-driven approach to supply chain analytics. By integrating SQL-based KPI engineering, statistical forecasting, and interactive visualization, the analysis provides a comprehensive view of operational health and future demand.

The finding that **simpler forecasting models often outperform complex alternatives** underscores the importance of empirical model selection. The identification of **Fashion as the highest-priority category for Q1 2026** provides a clear, actionable recommendation for inventory planning.

While limitations exist—particularly regarding data granularity and external factors—the framework established here provides a robust foundation for ongoing demand planning and supply chain optimization.

Project Deliverable

The project delivers a reproducible, scalable analytics solution that can be extended and refined as additional data and business requirements emerge.

A Key SQL Queries

A.1 Executive Summary Query

```sql
SELECT
    COUNT(*) AS records,
    SUM(units_sold) AS units_sold,
    ROUND(SUM(revenue_usd), 2) AS revenue_usd,
    ROUND(SUM(profit_usd), 2) AS profit_usd,
    ROUND(100.0 * SUM(profit_usd) / NULLIF(SUM(revenue_usd), 0), 2)
        AS weighted_profit_margin_pct,
    ROUND(AVG(lead_time_days), 2) AS avg_supplier_lead_days,
    ROUND(AVG(on_time_delivery_rate), 2) AS avg_on_time_delivery_pct
FROM supply_chain;
```

A.2 Category Financial Performance

```sql
SELECT
    product_category,
    SUM(units_sold) AS units_sold,
    ROUND(SUM(revenue_usd), 2) AS revenue_usd,
    ROUND(SUM(profit_usd), 2) AS profit_usd,
    ROUND(100.0 * SUM(profit_usd) / NULLIF(SUM(revenue_usd), 0), 2)
        AS weighted_margin_pct
FROM supply_chain
GROUP BY product_category
ORDER BY revenue_usd DESC;
```

A.3 Supplier Tier Classification

```sql
WITH suppliers AS (
    SELECT
        supplier_id,
        AVG(supplier_rating) AS avg_rating,
        AVG(lead_time_days) AS avg_lead_days,
        SUM(revenue_usd) AS revenue_usd
    FROM supply_chain
    GROUP BY supplier_id
),
tiers AS (
    SELECT
        *,
        CASE
            WHEN avg_rating >= 3.5 THEN 'Premium'
            WHEN avg_rating >= 2 THEN 'Standard'
            ELSE 'Low performer'
        END AS supplier_tier
    FROM suppliers
)
SELECT
    supplier_tier,
    COUNT(*) AS suppliers,
    ROUND(AVG(avg_lead_days), 2) AS avg_lead_days,
    ROUND(SUM(revenue_usd), 2) AS associated_revenue_usd
FROM tiers
GROUP BY supplier_tier
ORDER BY associated_revenue_usd DESC;
```

B Forecasting Model Evaluation Results

Table 11: Rolling-Origin Backtest Results by Category and Model

Category	Model	Mean MAE	Mean RMSE	Mean WAPE (%)
Electronics	Holt Trend	77,544	86,235	2.88
	Holt-Winters	85,106	88,617	3.14
	Seasonal Naive	99,329	110,136	3.66
	Naive	103,902	109,413	3.83
Fashion	Holt Trend	71,626	82,070	2.54
	Naive	88,903	102,973	3.15
	Seasonal Naive	128,831	138,730	4.56
	Holt-Winters	129,366	138,466	4.57
Food	Holt Trend	74,454	91,758	2.70
	Holt-Winters	113,581	132,368	4.08
	Naive	125,415	137,353	4.54
	Seasonal Naive	154,135	193,815	5.53
Health	Seasonal Naive	65,825	80,713	2.39
	Holt-Winters	75,559	95,252	2.74
	Holt Trend	100,470	116,673	3.64
	Naive	127,579	141,706	4.62
Home	Seasonal Naive	73,764	86,648	2.70
	Holt-Winters	111,936	131,817	4.04
	Holt Trend	145,039	159,402	5.27
	Naive	150,654	177,753	5.46

C Dashboard Screenshots

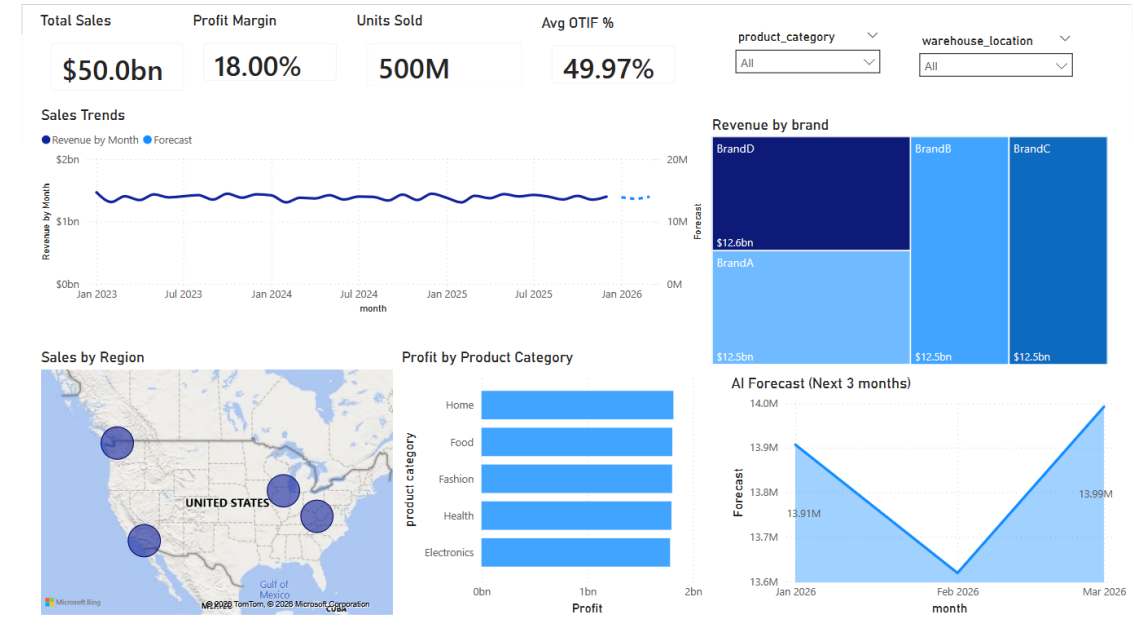


Figure 2: Executive Summary Dashboard

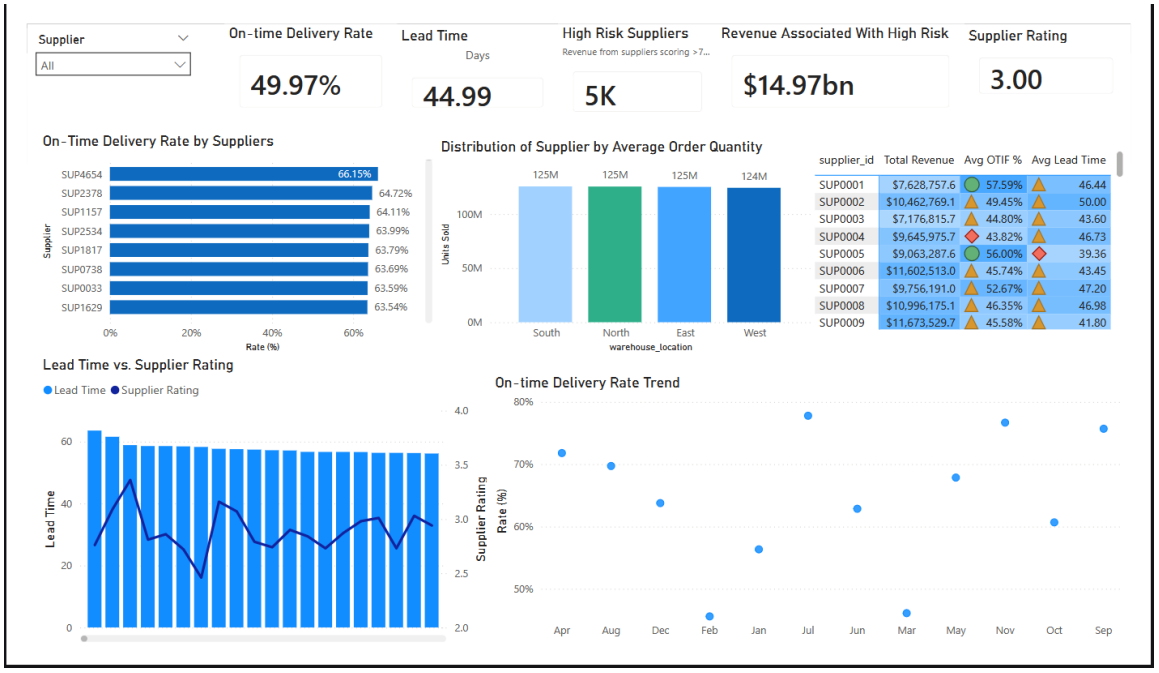


Figure 3: Supplier Performance Dashboard

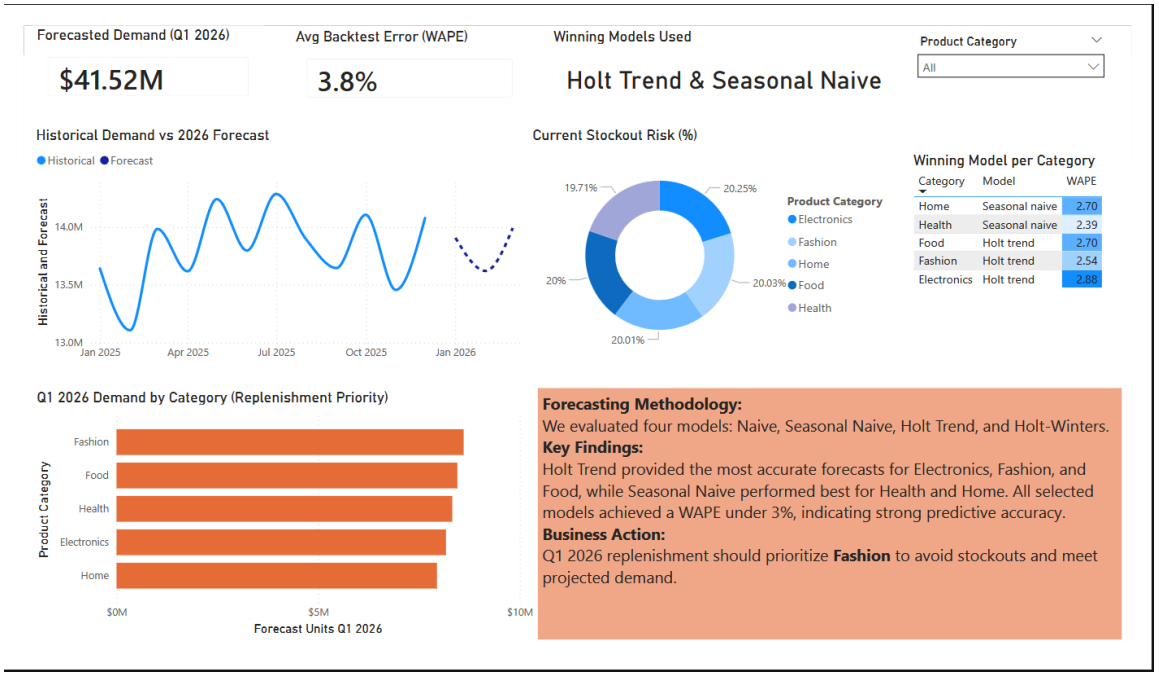


Figure 4: Demand Forecasting Dashboard